Motion as Communication: Plan-Conditioned Diffusion World Models for Negotiation at Unprotected Conflict Points

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***Abstract*—At an unprotected conflict point (e.g., an intersection, merging area, or roundabout), a driver does not merely predict what others will do; by moving, they influence what others will do. Standard autonomous-driving pipelines break this loop, treating surrounding agents as an exogenous stochastic process to be forecast and then avoided. We formalise the loop as an *influence channel*: a conditional kernel through which the ego's realised motion shifts the latent intentions of the agents around it. We then build the representation the channel requires, a *plan-conditioned* diffusion world model $p_\phi(y \mid h, u)$ that predicts the joint response of neighbouring vehicles to a *candidate* ego plan and can therefore be queried counterfactually. A reinforcement-learning agent consumes a compact summary of these queries that separates ordinary conflict geometry from plan-induced neighbour motion via predicted waypoints and an influence feature $\Delta(u)$ that is identically zero unless the forecast of y itself moves with the probe. We give a quantitative bound showing that the return attributable to influence is controlled by the total-variation sensitivity of the channel to the ego's plan. On a simulator with a known ground-truth channel, the intention head recovers the channel's direction (and most of its total variation on the merge). Closed-loop, five belief arms (none, geometry, kernel, history, diffusion) are compared under matched plan-as-action rollouts. Plan-conditioned diffusion significantly outperforms geometry and history on the merge; on the crossing and roundabout it matches geometry but does not significantly beat it. The channel is designed, not estimated from naturalistic data. The code and pre-trained models are available at: https://github.com/pebeigi/MaC-PDWM.**

***Index Terms*—Autonomous Driving, Diffusion Models, Multi-Agent Interaction, Reinforcement Learning, World Models**

I. Introduction

Merging into dense traffic or crossing an unsignalised intersection is not a prediction problem followed by a control problem. A driver who edges forward is making a claim; a driver who lifts off the throttle is making an offer. The other party reads the motion and responds, and the response was caused by the motion. Treating the surrounding traffic as an exogenous process to be forecast discards precisely the mechanism that resolves these situations, and the resulting agents are known to be either timid (waiting for a gap that a human would have created) or reckless, forcing right of way because nothing in the objective charges them for it.

This paper takes the causal loop seriously. Our starting point is that the latent intention of a neighbouring driver is not a fixed hidden parameter to be inferred, but a variable whose distribution the ego partially controls. We write this as an *influence kernel* $\rho(\theta^i \mid \tau^0, x^i)$ mapping the ego's realised trajectory to a distribution over a neighbour's behavioural type. When ρ does not depend on τ^0, the setting collapses to the familiar hidden-parameter POMDP and prediction-then-planning is recovered. When it does, the ego is not choosing a response to a future; it is choosing among futures.

This motivates the use of a plan-conditioned model. A predictor of the form $p(y \mid h)$, conditioned on history alone, cannot express influence at all: it has no argument through which a candidate plan could enter. We therefore learn a *plan-conditioned* generative model $p_\phi(y \mid h, u)$ over the joint future of the neighbours given a candidate ego plan u, using a conditional diffusion model so that the multimodality that matters here (e.g., one driver yields, another does not) is represented as distinct hypotheses rather than averaged into a trajectory nobody would drive.

The thing a history-based predictor plus a standard RL planner cannot represent is the interventional query $p(y \mid h, \mathrm{do}(u))$: "what would the neighbours do if I executed this plan?" That is the object we learn. These claims require three distinct evaluations, which structure Section VII: (Q1) can the model predict surrounding vehicles; (Q2) given the same scene history, does changing the candidate ego plan produce the corresponding change in predicted neighbour response; and (Q3) do these plan-conditioned predictions improve closed-loop decisions beyond a history-only world model?

We make four contributions.

1) **A decomposition.** We define the Negotiation-POMDP (Section II). This is not a new formalism: it is an ordinary POMDP with a partitioned state. The contribution is a decomposition that names the influence channel and thereby tells us what a model must condition on. Section II states precisely what the decomposition buys and what it does not.

2) **A measurable bound on the value of negotiation.** Proposition 2 splits the gain of a plan into an ordinary planning term and an influence term, and bounds the influence term by $R_{\max} \cdot \mathrm{TV}(\rho(\cdot \mid u), \rho(\cdot \mid u_0))$. Severing the type channel sets the latent-type contribution to negotiation gain to zero. Any remaining advantage of $p(y \mid h, u)$ over $p(y \mid h)$ can therefore arise only from other action-dependent interaction mechanisms, such as ordinary kinematic or car-following responses, rather than from changes in the latent type distribution.
3) **A plan-conditioned diffusion world model with a residual target** (Section IV), together with an evaluation protocol that compares best-of-S generative metrics against a stochastic baseline using the same sample budget.
4) **An identification argument for the counterfactual query, with its limits.** Randomising the behaviour policy (gap ϵ-noise, constant accel, open-loop F-step plans) licenses $p(y \mid h, \mathrm{do}(u))$ on the support of that mixture. Channel recovery checks *direction* against a known kernel; it does not identify ρ or prove that the closed-loop policy exploits influence rather than risk. The belief interface is constructed so that a geometry-only encoder cannot fake the influence feature (Section V); a remaining closed-loop tie then cannot be blamed on a risk-only bottleneck. If the `kernel` arm also fails to beat `geometry`, this suggests that the interaction regime itself may provide limited additional value beyond geometric risk information, rather than the observed tie being attributable solely to world-model error.

II. The Negotiation-POMDP

A. Setup

At decision step t (interval $\Delta = 0.4\,\mathrm{s}$) the scene contains an ego vehicle, indexed 0, and N_t surrounding vehicles. Table VII in Appendix fixes notation and ties each object to the released implementation. All quantities are expressed in the ego frame at time t, so the policy and the world model see relative interaction geometry rather than map coordinates.

B. Definition and what it does and does not claim

Definition 1 (Negotiation-POMDP): A Negotiation-POMDP is the tuple $\mathcal{M} = \langle \mathcal{S}, \mathcal{A}, \Omega, \mathcal{T}, \mathcal{O}, \Theta, \rho, r, \gamma \rangle$ with state $s_t = (x_t^0, x_t^{1:N_t}, \theta^{1:N_t})$, observation kernel $\mathcal{O}(o_t \mid s_t)$ emitting only kinematic quantities, and an *influence kernel*

$$\rho\big(\theta^i \,\big|\, \tau^0_{t-H:t},\, x_t^i\big), \tag{1}$$

under which the latent type of vehicle i is drawn as a function of the ego's own realised trajectory $\tau^0_{t-H:t}$ at the moment i commits to a decision.

Two consequences follow. First, θ is hidden, so the ego must maintain a belief $b_t(\theta) = \mathbb{P}(\theta \mid o_{0:t}, a_{0:t-1})$. Second, and this is the point, because ρ depends on τ^0 the ego's action changes the distribution it will later face: it does not merely predict the future, it partially selects it.

Remark 1 (What Definition 1 is not): Definition 1 does not describe a class of problems outside the standard POMDP. A POMDP whose state contains θ and whose transition kernel depends on recent action history is an ordinary POMDP, and any Negotiation-POMDP can be written as one. The content of the definition is a *decomposition*: it separates the state into kinematics the ego observes and types it does not, and it names the specific conditional dependence (of type on ego trajectory) that a model must represent. This is a modelling prescription, not an expressiveness claim. Its value is that it identifies u as a necessary argument of the world model, and Proposition 2 shows the resulting quantity is measurable. When ρ is constant in τ^0, Definition 1 reduces to a hidden-parameter POMDP [1] and prediction-then-planning is optimal.

We take $\Theta = \{\textsc{yield}, \textsc{contest}\}$, and we define the label by the *realised* decision at the conflict point rather than by the sampled type. A vehicle that never reaches the decision zone is unlabelled and masked out of every loss: a driver that has not yet had to decide is genuinely ambiguous, and the model should be permitted to say so.

III. A simulator with a known channel

We instantiate three unprotected conflicts in SUMO [2]: a four-way unsignalised crossing, a highway merge, and a roundabout (Fig. 1). On the crossing the ego traverses south-to-north under direct control with SUMO's safety overrides disabled, so right of way must be resolved by the policy rather than by the simulator. Background traffic crosses east–west with Poisson arrivals. Qualitative examples are deferred to Appendix B (Figs. 6, 7, and 8).

Each background vehicle is assigned a hidden type at insertion: a *yielder* always concedes, a *contester* never does, and a *reactive* driver decides on approach. Types are drawn with probabilities $(0.1, 0.1, 0.8)$ over (yielder, contester, reactive), so most neighbours are influenceable. For a reactive driver at signed distance d^i from the conflict point, the concession probability is

$$\mathbb{P}(\theta^i = \textsc{yield} \mid \tau^0) = \sigma\big(\beta_1(\mathrm{TTC}^i - \mathrm{TTC}^0) + \beta_2\,\bar{a}^0 + \beta_0\big), \\ (\beta_1, \beta_2, \beta_0) = (0.3,\, 2.5,\, -0.3), \tag{2}$$

implemented in `SocialDriverManager._decide_yield`. Here $\bar{a}^0$ is a $0.8\,\mathrm{s}$ exponential moving average of the ego's *commanded* acceleration (decision interval $\Delta = 0.4\,\mathrm{s}$), and the driver commits once it is within $35\,\mathrm{m}$ of the conflict on the crossing and merge ($22\,\mathrm{m}$ on the roundabout, with an incoming-edge allowlist). An unobservable per-driver offset ("stubbornness") is drawn at insertion and added to the logit, so the commitment is not a deterministic function of quantities already visible in h. On commit, a concession also issues an explicit `slowDown` (and a contest holds or raises speed), so the interventional response is kinematic inside the F-step prediction horizon rather than a silent junction-model switch. The two β terms have distinct meanings. The first is priority: a neighbour concedes more readily when the ego will plainly arrive first. The second is *communication*: an ego that accelerates is claiming the conflict point and raises the

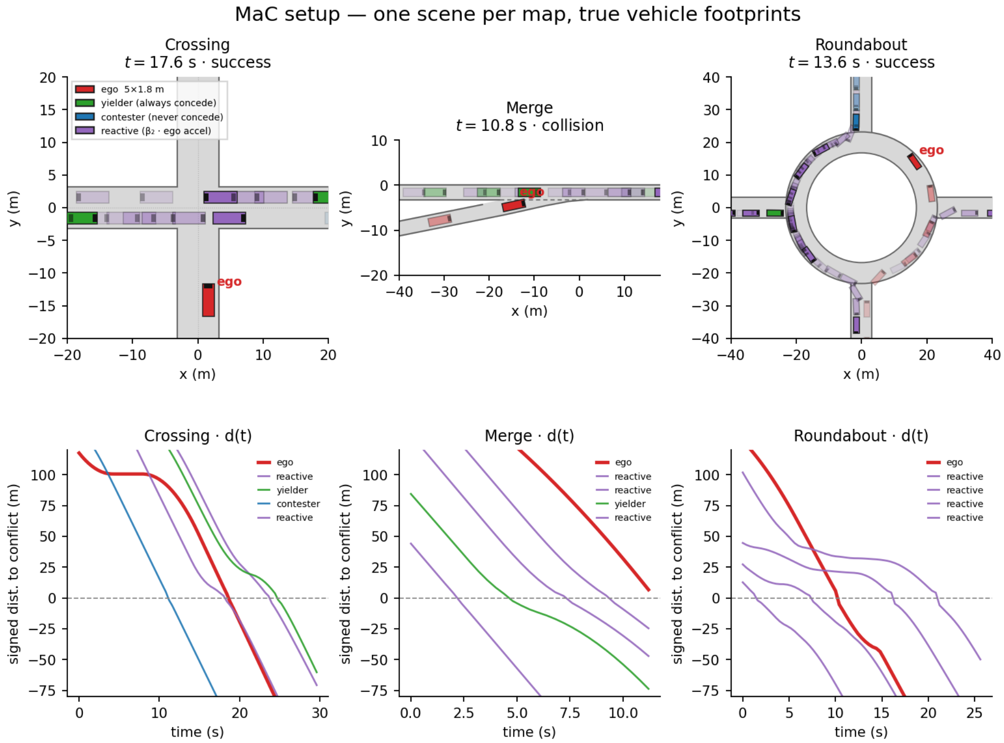


Fig. 1. Simulator setup. One snapshot per map (crossing, merge, roundabout) with $5{\times}1.8\,\mathrm{m}$ footprints.

probability that the other driver concedes, whereas an ego that decelerates is offering to give way and invites the other driver to take it. Setting $\beta_2 = 0$ severs the *type*-channel while leaving geometry, type priors, and traffic density untouched. SUMO car-following still reacts to the ego, so $\beta_2 = 0$ is not "no interaction": it is "latent type independent of ego motion." That is the control in Section VII-D.

The channel is known to the simulator and never exposed to the agent, which is what lets us ask whether a learned model recovers it. We are explicit in Section VIII that this is a designed channel, not one estimated from naturalistic driving.

a) Reward.: Writing n_t^{brake} for the number of hard braking events the ego *causes* in others,

$$\begin{aligned} r_t = {} & \underbrace{w_p \tfrac{v_t}{v_{\max}} - w_\tau}_{\text{progress}} - \underbrace{w_c \mathbb{1}\{\text{collision}\}}_{\text{safety}} \\ & + \underbrace{w_g \mathbb{1}\{\text{arrival}\}}_{\text{goal}} - \underbrace{w_{\mathrm{to}} \mathbb{1}\{\text{timeout}\}}_{\text{liveness}} \\ & - \underbrace{w_a a_t^2 + w_j (\dot{a}_t/10)^2}_{\text{comfort}} \\ & - \underbrace{w_x (\delta_x - g_t)^+}_{\text{proximity}} - \underbrace{w_s\, n_t^{\mathrm{brake}}}_{\text{social}}, \end{aligned} \tag{3}$$

with $(w_p, w_\tau, w_c, w_g, w_{\mathrm{to}}, w_a, w_j, w_x, w_s) = (0.2, 0.06, 20, 20, 15, 0.005, 0.01, 0.05, 0.15)$ and $\delta_x = 8\,\mathrm{m}$. The elevated time penalty ($w_\tau{=}0.06$) makes waiting for a natural gap costly relative to the success bonus, so negotiation can pay for itself. The jerk term is scaled by 10 so that typical decision-to-decision accel changes do not dominate progress. Further discussion of the social and liveness terms is in Appendix C.

IV. Plan-Conditioned Diffusion World Model

a) Residual target.: Background traffic is close to constant velocity, so predicting absolute motion spends capacity on the trivial component. We predict the residual from a constant-velocity roll-out,

$$\begin{aligned} y_{t,f}^i &= \frac{\left(p_{t+f}^i - p_t^i\right) - v_t^i\, f\Delta}{\lambda}, \quad f = 1, \ldots, F, \\ \lambda &= 8\,\mathrm{m}. \end{aligned} \tag{4}$$

All behaviour of interest (e.g., braking to yield, holding speed to contest) lives in y.

b) Objective.: Let $c_t = \mathrm{Enc}_\phi(h_t, u_t)$ be the context and let m_f^i mask vehicles absent at step f. With a cosine schedule $\bar{\alpha}_k$ and $y_k = \sqrt{\bar{\alpha}_k}\, y_0 + \sqrt{1-\bar{\alpha}_k}\,\varepsilon$, the model minimises

$$\begin{aligned} \mathcal{L}(\phi) = \mathbb{E}_{k,\varepsilon,(h,u,y)} & \left[\frac{\sum_{i,f} m_f^i \|\varepsilon - \varepsilon_\phi(y_k, k, c)\|^2}{\sum_{i,f} m_f^i} \right] \\ & + \eta\, \mathbb{E}\big[\mathcal{H}\big(\theta^i,\, q_\phi(\cdot \mid h, u)\big)\big], \quad \eta = 1.0, \end{aligned} \tag{5}$$

where $q_\phi(\theta^i \mid h_t, u_t)$ is an auxiliary intention head sharing the encoder. The default η is 1.0; on the crossing and roundabout the trained models use a larger intention weight (Appendix B). Note that q_ϕ estimates a *plan-conditioned* belief (e.g., the probability that neighbour i concedes *if the ego executes* u_t) which is a counterfactual query against the channel, not a passive estimate. Cross-entropy is restricted to open-loop (constant / hold / plan) episodes, so the head is fitted to $p(\theta \mid h, \mathrm{do}(u))$ rather than the observational conditional in which u is a function of h. Training also drops the plan with probability 0.1, which gives the encoder a well-defined history-only branch.

The essential structural choice is that c_t depends on u_t. The model is therefore controllable, $p_\phi(y \mid h, u)$, and can be queried counterfactually. Sampling uses a strided DDIM schedule with x_0 clipping, which keeps a query cheap enough to run inside the RL loop. At query time the intention logits are written as a history-only term plus a plan residual, and the residual is scaled by a guidance weight w calibrated on held-out open-loop cross-entropy and stored in the checkpoint (w=1 is the plain conditional).

c) Identification: why the counterfactual query is licensed.: Identification is achieved through exogenous action variation in the data collection policy; details are provided in Appendix D.

V. Negotiation-aware planning

a) Belief features.: The closed-loop policy uses `plan_action`: the discrete action is itself a probe from $\mathcal{U} = \{-4, 0, +3\}$ (yield / hold / assert), held for `commit_steps` $= 10$ environment steps (4 s), shorter than the F=20 query horizon so the policy can still wait out a hard contester. For each $u \in \mathcal{U}$ the encoder draws S futures $y^{(1:S)} \sim p_\phi(\cdot \mid h_t, u)$ with deterministic DDIM (η=0) and a shared latent across the probe batch so differences isolate u, and forms

$$g_s(u) = \min_{f,i} \left\| p^0_f(u) - \hat{p}^{i,(s)}_f \right\|, \tag{6}$$

the closest approach between the probe ego trajectory and predicted neighbours (the K nearest within a radius, not an oracle of who is about to commit). Risk, margin and spread of $g_s(u)$ can be produced by ego geometry alone when a constant-velocity neighbour forecast already moves them when the probe accelerates so they do not identify negotiation. Learned arms therefore reuse the same deterministic CV core stats for risk/clearance and add residual features the CV baseline cannot copy. The encoder also writes, for each u, (i) the predicted closest-neighbour position at mid- and end-horizon and (ii) the mean displacement of predicted neighbours relative to the hold probe,

$$\Delta(u) = \frac{1}{|\mathcal{N}|} \sum_{i \in \mathcal{N}} \left\| \bar{p}^i(u) - \bar{p}^i(0) \right\|, \tag{7}$$

which is identically zero unless the forecast of y itself moves with u. A `geometry` or `history` encoder has $\Delta \equiv 0$; only a plan-conditioned model can make it nonzero. Details are in Appendix E.

The policy $\pi_\vartheta(a_t \mid o_t, \psi_t)$ maximises $J(\vartheta) = \mathbb{E}_{\pi_\vartheta}[\sum_t \gamma^t r_t]$ with $\theta^i \sim \rho(\cdot \mid \tau^0)$, using clipped PPO [3] with a running observation normalisation: raw kinematics and belief features differ by two orders of magnitude, and without per-feature scaling a small-magnitude slot such as $\Delta(u)$ is effectively absent from a tanh trunk. In the simulator J is a negotiation objective. Probe risk can still support ordinary risk-aware planning, which a history-only or geometry encoder already provides. What those encoders *cannot* provide is a nonzero $\Delta(u)$. A closed-loop diffusion–geometry gap under this interface cannot be explained as "the policy only saw a collision margin"; the absence of such a gap cannot be blamed on that bottleneck either. Section VII-C–VII-D report which of those two readings the data support.

VI. What can be proved

The following results characterize the theoretical properties of the proposed formulation. We show that point-forecast planning can be suboptimal in the considered setting and bound the influence contribution by the sensitivity of the channel.

Proposition 1 (Certainty-equivalence regret): Consider one decision with binary latent $\theta \in \{Y, C\}$, $\mathbb{P}(\theta = Y) = p$, and actions $\{go, wait\}$ with $c(go, Y) = 0$, $c(go, C) = C > 0$, $c(wait, \cdot) = d > 0$. A distributional planner attains $\min\{(1-p)C, d\}$. A certainty-equivalence planner that replaces θ by its most likely value and acts optimally for it chooses *go* whenever $p > 1/2$, incurring true expected cost $(1-p)C$. Its regret is $\mathcal{R} = \max\{0, (1-p)C - d\}$, strictly positive whenever $d < (1-p)C$.

Proof: Immediate by comparing the two expected costs. □

The regime $d < (1-p)C$ is precisely an unprotected crossing: waiting is cheap, collisions are expensive, and residual contest probability is far from zero. Proposition 1 also explains why a *mean-trajectory* forecast is worse than a mode: averaging a yielding and a contesting trajectory produces a path no driver would execute, so the planner optimises against a future of probability zero. We stress that the proposition is a statement about certainty equivalence in a one-shot decision; it explains why the failure mode exists, and does not bound the RL agent's regret.

The next result is the one that carries weight, because it makes the value of negotiation measurable rather than rhetorical.

Proposition 2 (Negotiation gain is bounded by channel sensitivity): Let $R(u, \theta) \in [0, R_{\max}]$ and $J(u) = \mathbb{E}_{\theta \sim \rho(\cdot|u)}[R(u, \theta)]$. For any plans u, u_0, decompose

$$\begin{aligned} J(u) - J(u_0) = &\underbrace{\mathbb{E}_{\theta \sim \rho(\cdot|u_0)}\big[R(u,\theta) - R(u_0,\theta)\big]}_{G_{\text{plan}}} \\ &+ \underbrace{\mathbb{E}_{\theta \sim \rho(\cdot|u)}[R(u,\theta)] - \mathbb{E}_{\theta \sim \rho(\cdot|u_0)}[R(u,\theta)]}_{G_{\text{infl}}}. \end{aligned} \tag{8}$$

Then the influence term obeys

$$\left|G_{\text{infl}}\right| \;\leq\; R_{\max} \cdot \mathrm{TV}\big(\rho(\cdot \mid u),\, \rho(\cdot \mid u_0)\big), \tag{9}$$

and $G_{\text{infl}} = 0$ for every u whenever ρ does not depend on u. For binary Θ the bound is $|G_{\text{infl}}| \leq R_{\max} \, |\mathbb{P}(\text{YIELD} \mid u) - \mathbb{P}(\text{YIELD} \mid u_0)|$.

Proof: The decomposition is an addition and subtraction of $\mathbb{E}_{\rho(\cdot|u_0)}[R(u,\theta)]$. For the bound, $G_{\text{infl}} = \int R(u,\theta)\,(\rho(d\theta \mid u) - \rho(d\theta \mid u_0))$, and for any f with range in an interval of length L, $|\int f\, d(P-Q)| \le L \cdot \mathrm{TV}(P,Q)$; take $L = R_{\max}$. If $\rho(\cdot \mid u) = \rho(\cdot \mid u_0)$ the integrand measure is zero. □

Proposition 2 is the formal content of "motion is communication", and unlike a feasibility argument it says something with consequences. The gain attributable to influence is *at most* the reward scale times how far the ego can move the type distribution, so an environment with a weak channel cannot reward negotiation however good the model is. It also yields a falsifiable prediction: setting $\beta_2 = 0$ in Eq. (2) forces $\mathrm{TV} = 0$, hence $G_{\text{infl}} = 0$, and any advantage of a plan-conditioned model over a history-only one must disappear. Section VII-D tests this. In our simulator the kernel is known, so the bound can be evaluated: at zero priority margin the probe set spans $\mathbb{P}(\textsc{yield})$ from 0.016 to 0.954 under the true kernel, giving $\mathrm{TV} = 0.938$.

Remark 2 (Resolution of the sampled risk estimate): $\hat{P}_S(u) = \frac{1}{S}\sum_s \mathbb{1}\{g_s(u) < \delta\}$ is unbiased for the conflict probability under p_ϕ, and by Hoeffding $\mathbb{P}(|\hat{P}_S - P| \ge \epsilon) \le 2\exp(-2S\epsilon^2)$. At $S = 8$ the 95% resolution is only $\epsilon \approx 0.48$, so a single-step risk estimate is coarse; the policy sees a temporal sequence of such estimates, which mitigates this. Because S is therefore a first-order design parameter rather than an incidental hyperparameter, we sweep it in Section VII-E.

VII. Experiments

We test three claims separately, because a single closed-loop number cannot tell them apart.

- **Q1 Prediction.** Can $p_\phi(y \mid h, u)$ forecast neighbours, and how does it compare to constant velocity and to $p(y \mid h)$ (§VII-A)?
- **Q2 Counterfactual prediction.** Given the same h, does changing u move the forecast in the direction of the simulator's $\mathrm{do}(u)$ response (§VII-B)?
- **Q3 Useful negotiation.** Does giving those queries to an RL planner improve decisions beyond a history-only world model, and does that extra gain vanish when the type-channel is off (§VII-C–VII-D)?

a) Protocol.: The operating point matches the released pipeline: channel Eq. (2), type mix $(0.1, 0.1, 0.8)$, decision distance 35 m on the crossing and merge (22 m on the roundabout), intent window 0.8 s, prediction horizon F=20 (8 s), probes $\{-4, 0, +3\}$, and five belief arms (`none`, `geometry`, `kernel`, `history`, `diffusion`) on each of the three scenes, up to 10 seeds, 60 PPO iterations, with `plan_action` and `commit_steps`=10 (4 s). 10,201 crossing episodes (plus merge/roundabout, collected the same way) yield 347,823 training and 36,888 validation samples on the crossing. The split is over paired-intervention groups / episodes, not individual overlapping samples. Consecutive decision steps within an episode produce near-identical windows, so a random sample split would inflate validation metrics.

A. Q1: Prediction

Table I reports displacement errors. The comparison protocol matters more than the numbers. minFDE is a best-of-S statistic and it is not meaningful to compare it against a single deterministic forecast: doing so credits the generative model for sample diversity without charging it for calibration. We therefore report a constant-velocity roll-out perturbed by Gaussian noise at the empirical residual scale (11.40 m) with the same $S = 8$ budget, alongside the deterministic roll-out, and we report meanFDE as well as minFDE. Against that protocol minFDE is a win — 10.57 m versus 16.04 m for the matched stochastic baseline. The same comparison on all three maps is in Appendix B (Fig. 9). MeanFDE (25.43 m) exceeds deterministic constant velocity (21.71 m) on the crossing, as expected when a multi-hypothesis model spreads mass over modes that a single CV roll-out never visits. Plan-conditioning on the *logged* u is a weak test of Q2: if zeroing u at eval time barely changes FDE, u was not needed to predict the realised future. We report that ablation in Table I (9.49 m with the plan zeroed vs 10.57 m with the plan; history-only 8.87 m).

B. Q2: Counterfactual prediction

Q2 is not "FDE on logged (h, u, y)." It is: same h, different u, different y. Table II and Fig. 2 probe the intention head on the crossing; Fig. 10 in Appendix B repeats the probe on all three maps. A model that had memorised the marginal would show no probe dependence, and one that had learned a spurious correlate could show the reverse. Trajectory shift is reported against sample spread: a probe effect smaller than the model's own noise is not usable. An interventional replay of the same scene under $\mathrm{do}(u)$ (Appendix B, Table IX and Fig. 12) scores forecasts against realised neighbour futures; a history-only model must predict nearly the same y for every u.

Probe TV collapses if averaged over all neighbours; we report the deciding population (Appendix B). Even there, recovered TV is weaker than the kernel and trajectory shifts sit near sample spread.

C. Q3: Useful negotiation

Q3 asks whether counterfactual queries improve *decisions* beyond $p(y \mid h)$+RL. A fair test has to survive a specific objection: the policy never sees the sample tensor, only a summary, and a summary built from $g_s(u)$ (6) can be reproduced by ego geometry alone. The encoder of Section V answers that objection by adding waypoints and $\Delta(u)$ (7). Geometry and history then have $\Delta \equiv 0$; a remaining diffusion–geometry gap cannot be "the policy only saw a smaller collision margin," and a remaining tie cannot be blamed on that bottleneck either. The `kernel` arm is the engineered channel baseline: it gives the policy an analytic approximation of ρ on top of the same geometry. A diffusion–geometry gap that `kernel` also matches is consistent with influence; a gap that history also matches is not.

Table III reports the crossing; Table IV the merge; Table V the roundabout; Figs. 3 and 4 summarize the same numbers.

TABLE I

DISPLACEMENT ERROR ON HELD-OUT CROSSING EPISODES, IN METRES, $K = 5$ NEIGHBOURS OVER $F = 20$ STEPS (8 S). MINFDE IS A BEST-OF-S STATISTIC AND IS ONLY COMPARABLE ACROSS ROWS WITH THE SAME S; THE DETERMINISTIC CONSTANT-VELOCITY ROLL-OUT IS THE REFERENCE FOR MEANFDE, AND THE NOISE-PERTURBED ROLL-OUT WITH A MATCHED $S = 8$ BUDGET IS THE REFERENCE FOR MINFDE.

Model	S	**minFDE**	**meanFDE**	**minADE**	**meanADE**
Constant velocity	1	21.71	21.71	9.12	9.12
Constant velocity + noise	8	16.04	28.58	16.37	19.02
Diffusion $p(y \mid h, u)$ (DDIM T=25)	8	10.57	25.43	4.21	10.91
Diffusion $p(y \mid h)$ (plan zeroed)	8	9.49	25.39	3.68	11.03
History-only WM $p(y \mid h)$	8	8.87	21.59	3.22	9.10

TABLE II

CHANNEL RECOVERY ON THE CROSSING. THE MODEL IS PROBED WITH $\{-4, 0, +3\}$; THE SIMULATOR'S KERNEL REQUIRES $\Pr(\text{YIELD})$ TO INCREASE WITH EGO ACCELERATION. TRAJECTORY SHIFT IS THE MEAN DISPLACEMENT OF THE PREDICTED FUTURES RELATIVE TO THE HOLD PROBE, TO BE READ AGAINST THE SPREAD ACROSS DIFFUSION SAMPLES (4.72 M). GROUND TRUTH USES THE HELD-OUT SCENE'S TTC MARGIN (SCENE-CONDITIONED).

Probe u	a **(m/s^2)**	$\Pr(\text{YIELD})$ **dec.**	$\Pr(\text{YIELD})$ **all**	$\Pr(\text{YIELD})$ **truth**	**Shift (m)**
yield	-4	0.243	0.330	0.016	1.38
hold	0	0.352	0.445	0.361	—
assert	$+3$	0.918	0.790	0.954	2.49
Total variation		0.675	0.460	0.938	
Monotone in a		yes	yes	yes	

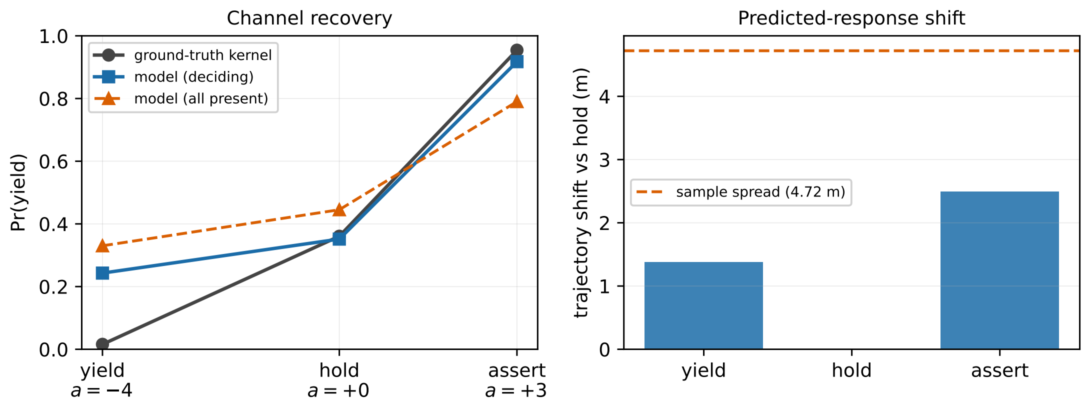


Fig. 2. Left: probe response of the learned intention head against the simulator's influence kernel. Right: mean trajectory shift of predicted futures relative to the hold probe, compared to the model's own sample spread (dashed).

In the Appendix, Fig. 13 shows the crossing learning curves, and Fig. 14 shows all three maps.

The comparison that matters is `none` versus `geometry` versus `kernel` versus `history` versus `diffusion`. A large `none`→`geometry` lift with `history`≈`diffusion` is risk-aware planning, not negotiation. A remaining diffusion–history *and* diffusion–geometry gap on return, with paired bootstrap CIs excluding zero, is the evidence Q3 requires.

On the **merge**, diffusion returns 16.02±2.01 versus geometry 10.43±4.28 and history 11.87±2.52 (paired Δ return vs geometry +4.87, 95% CI $[+2.37, +7.99]$, n=8). History does not significantly beat geometry. Merge seed counts are incomplete (n=7–9 per arm; paired comparisons use the overlapping seeds). These results suggest that the gain is related to plan conditioning rather than simply the use of a learned world model.

On the **crossing**, diffusion returns 11.97±1.49 versus geometry 11.60±1.43 (paired Δ return +0.36, 95% CI $[-0.60, +1.31]$, n=10). The CI includes zero. History (11.51±1.40) and kernel (11.55±1.36) also match geometry; the lift over `none` (10.17±1.83) is therefore the geometry features, not plan-conditioned neighbour forecasts.

On the **roundabout**, kernel and history significantly beat geometry (paired Δ return +1.74 and +1.78; both CIs exclude zero). Diffusion (11.46±1.81) matches geometry (11.55±1.16; paired -0.09, CI $[-1.48, +1.13]$ includes zero). An analytic channel feature helps in this scene; the learned $p(y \mid h, u)$ does not convert that into a closed-loop gain.

D. Severing the channel

Proposition 2 predicts that with $\beta_2 = 0$ the type-influence term is identically zero. SUMO car-following still reacts, so a residual gap of any world model over `none` is ordinary prediction (including kinematic interaction) and is expected. What the proposition forbids is an *additional* gain of $p(y \mid h, u)$ over $p(y \mid h)$. Table VI is the matched $\beta_2 = 0$ control on the crossing (Fig. 5). The merge diffusion–history gap under

TABLE III

CLOSED-LOOP PERFORMANCE ON THE CROSSING (COMMIT_STEPS= 10, PLAN-AS-ACTION, $F = 20$), MEAN $\pm$ S.D. OVER n=10 SEEDS.

Belief	Success (%)	Collision (%)	Timeout (%)	Return	Crossing steps	Induced brakes
`none` (no world model)	65.00±4.49	35.00±4.49	0.00±0.00	10.17±1.83	58.91±1.50	0.00±0.00
`geometry` (ego-plan CV risk)	69.00±3.61	31.00±3.61	0.00±0.00	11.60±1.43	60.80±3.68	0.00±0.00
`kernel` (analytic channel feature)	69.00±3.82	30.80±3.92	0.20±0.60	11.55±1.36	60.96±2.63	0.00±0.00
`history` (plan dropped)	69.20±3.71	30.80±3.71	0.00±0.00	11.51±1.40	64.12±2.29	0.00±0.00
`diffusion` (plan-conditioned)	70.00±4.00	30.00±4.00	0.00±0.00	11.97±1.49	62.13±2.97	0.00±0.00

TABLE IV

CLOSED-LOOP PERFORMANCE ON THE MERGE (COMMIT_STEPS= 10, PLAN-AS-ACTION, $F = 20$), MEAN $\pm$ S.D. OVER SEEDS. SEED COUNTS ARE INCOMPLETE: n=8 (NONE), 9 (GEOMETRY, HISTORY, DIFFUSION), 7 (KERNEL).

Belief	Success (%)	Collision (%)	Timeout (%)	Return	Crossing steps	Induced brakes
`none` (no world model)	74.25±5.04	25.25±3.86	0.50±1.32	13.64±2.45	76.35±1.16	0.00±0.00
`geometry` (ego-plan CV risk)	67.56±8.83	30.67±6.11	1.78±3.33	10.43±4.28	77.55±1.42	0.00±0.00
`kernel` (analytic channel feature)	70.29±3.61	28.57±3.16	1.14±0.99	11.64±1.60	77.41±1.73	0.00±0.00
`history` (plan dropped)	70.22±5.61	29.78±5.61	0.00±0.00	11.87±2.52	77.40±0.50	0.00±0.00
`diffusion` (plan-conditioned)	79.78±4.94	20.00±5.08	0.22±0.63	16.02±2.01	79.23±1.36	0.00±0.00

TABLE V

CLOSED-LOOP PERFORMANCE ON THE ROUNDABOUT (COMMIT_STEPS= 10, PLAN-AS-ACTION, $F = 20$), MEAN $\pm$ S.D. OVER n=10 SEEDS.

Belief	Success (%)	Collision (%)	Timeout (%)	Return	Crossing steps	Induced brakes
`none` (no world model)	68.20±3.94	31.80±3.94	0.00±0.00	9.89±1.61	77.57±3.86	0.00±0.00
`geometry` (ego-plan CV risk)	72.80±3.12	27.20±3.12	0.00±0.00	11.55±1.16	77.80±1.82	0.00±0.00
`kernel` (analytic channel feature)	76.80±4.66	23.20±4.66	0.00±0.00	13.28±1.84	76.36±1.48	0.00±0.00
`history` (plan dropped)	77.00±4.58	23.00±4.58	0.00±0.00	13.32±1.84	76.24±0.94	0.00±0.00
`diffusion` (plan-conditioned)	72.80±4.58	27.20±4.58	0.00±0.00	11.46±1.81	78.94±2.95	0.00±0.00

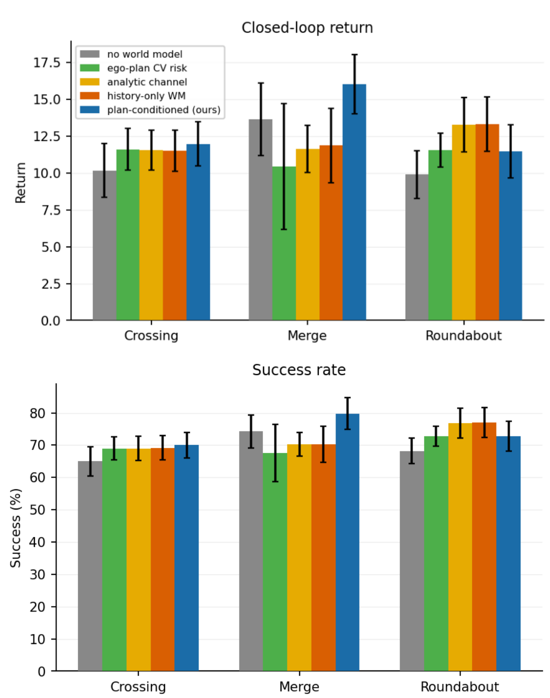


Fig. 3. Closed-loop return and success on all three maps (mean $\pm$ s.d. over seeds). Geometry captures most of the lift on the crossing; plan-conditioned diffusion is the outlier on the merge; kernel and history lead on the roundabout.

$\beta_2 > 0$ remains the primary Q3 evidence; we did not re-run the merge suite at $\beta_2 = 0$.

E. Sample budget

Remark 2 bounds the Monte-Carlo error of the *risk feature*, not PPO sample efficiency. Table VIII and Fig. 11 vary $S \in \{1, 4, 8, 16\}$. S=8 is the operating point used elsewhere.

VIII. LIMITATIONS

We state these plainly because several bear directly on how the results should be read.

- **Designed channel, favourable causal structure.** ρ is synthetic. Recovery and closed-loop gains do not show that the method discovers unknown negotiation in real traffic. $\beta_2 = 0$ still leaves SUMO car-following. Recovered model TV is much weaker than the designed kernel on the crossing and roundabout; on the merge it matches (0.624 vs 0.650).
- **Closed-loop gains are scene-dependent.** Geometry-only risk may capture most of the closed-loop lift under the 38-scalar interface. On the crossing, `kernel` does not beat `geometry`, suggesting that geometric risk captures most decision-relevant information. On the roundabout, `kernel` and `history` beat `geometry` while diffusion does not, indicating that useful interaction information is present but not fully exploited. The merge is the regime where influence is both identifiable and useful. Understanding why prediction yields a clear gain on the merge but not on the crossing or roundabout is an important direction for future work.
- **Weak counterfactual trajectory signal.** Probe intention can be directionally correct while trajectory shifts sit near

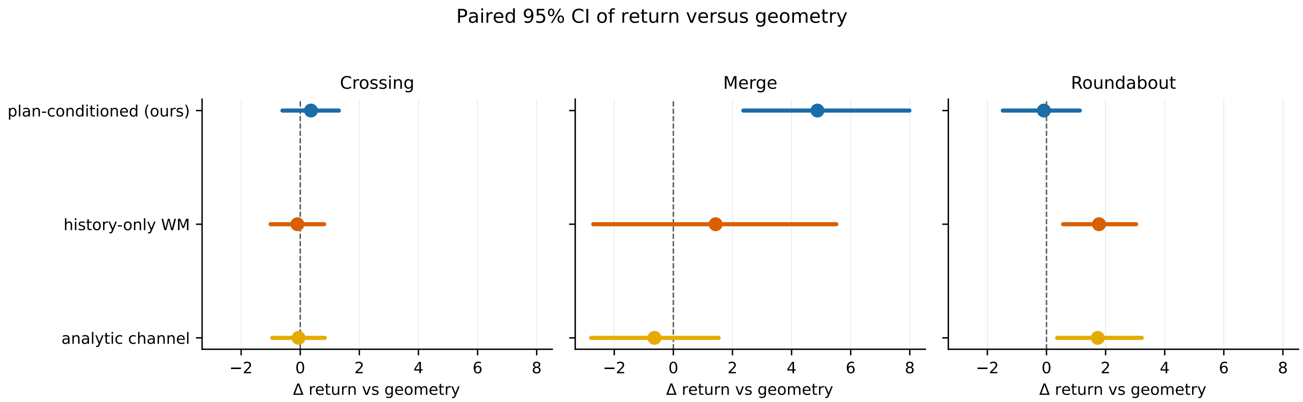


Fig. 4. Paired 95% CI of return versus the geometry arm, seed-matched. The CI excludes zero only for diffusion on the merge, and for kernel and history on the roundabout. Crossing CIs all include zero.

TABLE VI
TYPE-CHANNEL OFF ($\beta_2 = 0$). LATENT TYPES NO LONGER DEPEND ON EGO MOTION, SO PROPOSITION 2 FORCES $G_{\text{infl}} = 0$. SUMO CAR-FOLLOWING STILL REACTS TO THE EGO. MEAN ± S.D. OVER SEEDS.

Belief	**Success (%)**	**Collision (%)**	**Timeout (%)**	**Return**	**Crossing steps**	**Induced brakes**
`none` (no world model)	61.80±1.40	38.20±1.40	0.00±0.00	9.10±0.54	50.68±0.29	0.00±0.00
`history` (plan dropped)	61.80±2.09	38.00±2.37	0.20±0.60	9.00±0.83	50.84±0.42	0.00±0.00
`diffusion` (multi-hypothesis)	62.60±1.80	37.00±1.61	0.40±0.80	9.32±0.71	51.49±1.43	0.00±0.00

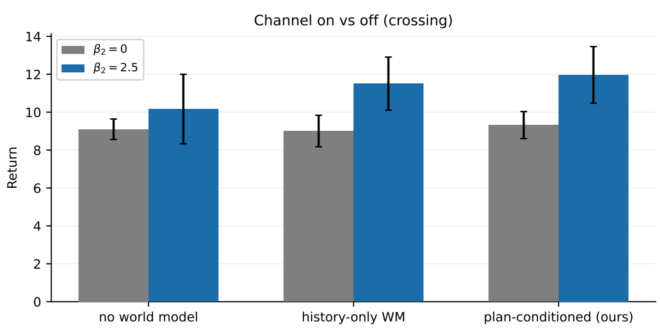


Fig. 5. Crossing return with the type-channel off (β_2=0) versus the operating point (β_2=2.5). At β_2=0 the three arms collapse. With the channel on, history and diffusion both lift over `none`.

sample spread. Table IX reports interventional FDE on n=22 matched crossing scenes: history has lower FDE with zero shift; diffusion moves with u. We do not claim a numerical CF FDE win.

- **Multimodality is regime-dependent.** On the crossing, S=1 return is close to S=8 (11.04 vs 11.97) but does not match it. We do not claim diffusion is universally better.
- **Coarse queries and a compact summary.** Three probes, K nearest neighbours, no oracle of who is deciding. The policy sees a 38-scalar summary that includes neighbour waypoints and the influence feature $\Delta(u)$, not the raw sample tensor. Identifying relevant vehicles in deployment is itself hard.
- **Limited shift and scenarios.** Results are in-family on three SUMO maps (crossing, merge, roundabout). No published predictor/negotiator baseline.
- **Theory scope.** Proposition 1 is one-shot. Proposition 2 is an upper bound. The Hoeffding result characterizes the finite-sample resolution of the risk estimator and should not be interpreted as a bound on PPO sample efficiency.

IX. CONCLUSION

We formalised negotiation at unprotected conflict points as an influence channel and bounded the return attributable to that channel by its total-variation sensitivity to the ego's plan. A plan-conditioned generative model represents this channel. The belief interface includes the plan-response feature $\Delta(u)$ and a `kernel` arm that injects an analytic channel feature on top of constant-velocity conflict risk. Empirically, the intention head recovers the channel's direction and, on the merge, most of its total variation. Closed-loop, plan-conditioned diffusion significantly outperforms geometry and history on the merge, while matching geometry on the crossing and roundabout. Collision reductions from any world-model or geometry-only belief should not alone be interpreted as negotiation; the relevant evidence is an additional diffusion–history gap beyond geometry, observed on the merge. This extra gap is absent on the crossing, where the three arms also collapse at $\beta_2 = 0$; the merge was not re-evaluated with the type channel removed. Whether a channel estimated from real drivers has sufficient sensitivity for Proposition 2 to matter remains open. Future work will investigate why plan-conditioned prediction provides a clear gain on the merge but not on the crossing or roundabout, including richer plan probes, improved response modelling, and scene-adaptive belief representations.

REFERENCES

[1] F. Doshi-Velez and G. Konidaris, "Hidden parameter markov decision processes: A semiparametric regression approach for discovering latent task parametrizations," in *International Joint Conference on Artificial Intelligence (IJCAI)*, 2016.

[2] P. A. Lopez, M. Behrisch, L. Bieker-Walz, J. Erdmann, Y.-P. Flötteröd, R. Hilbrich, L. Lücken, J. Rummel, P. Wagner, and E. Wießner, "Microscopic traffic simulation using sumo," in *IEEE International Conference on Intelligent Transportation Systems (ITSC)*, 2018.

[3] J. Schulman, F. Wolski, P. Dhariwal, A. Radford, and O. Klimov, "Proximal policy optimization algorithms," *arXiv preprint arXiv:1707.06347*, 2017.

[4] D. Sadigh, S. Sastry, S. A. Seshia, and A. D. Dragan, "Planning for autonomous cars that leverage effects on human actions," in *Robotics: Science and Systems (RSS)*, 2016.

[5] D. Sadigh, N. Landolfi, S. S. Sastry, S. A. Seshia, and A. D. Dragan, "Planning for cars that coordinate with people: Leveraging effects on human actions for planning and active information gathering over human internal state," *Autonomous Robots*, vol. 42, no. 7, pp. 1405–1426, 2018.

[6] T. Salzmann, B. Ivanovic, P. Chakravarty, and M. Pavone, "Trajectron++: Dynamically-feasible trajectory forecasting with heterogeneous data," in *European Conference on Computer Vision (ECCV)*, 2020.

[7] Y. Yuan, X. Weng, Y. Ou, and K. M. Kitani, "Agentformer: Agent-aware transformers for socio-temporal multi-agent forecasting," in *IEEE/CVF International Conference on Computer Vision (ICCV)*, 2021.

[8] S. Shi, L. Jiang, D. Dai, and B. Schiele, "Motion transformer with global intention localization and local movement refinement," in *Advances in Neural Information Processing Systems (NeurIPS)*, 2022.

[9] J. Ngiam, B. Caine, V. Vasudevan, Z. Zhang, H.-T. L. Chiang, J. Ling, R. Roelofs, A. Bewley, C. Liu, A. Venugopal *et al.*, "Scene transformer: A unified architecture for predicting future trajectories of multiple agents," in *International Conference on Learning Representations (ICLR)*, 2022.

[10] E. Tolstaya, R. Mahjourian, C. Downey, B. Vadarajan, B. Sapp, and D. Anguelov, "Identifying driver interactions via conditional behavior prediction," in *IEEE International Conference on Robotics and Automation (ICRA)*, 2021.

[11] N. Rhinehart, R. McAllister, K. Kitani, and S. Levine, "Precog: Prediction conditioned on goals in visual multi-agent settings," in *IEEE/CVF International Conference on Computer Vision (ICCV)*, 2019.

[12] M. Janner, Y. Du, J. B. Tenenbaum, and S. Levine, "Planning with diffusion for flexible behavior synthesis," in *International Conference on Machine Learning (ICML)*, 2022.

[13] A. Ajay, Y. Du, A. Gupta, J. B. Tenenbaum, T. S. Jaakkola, and P. Agrawal, "Is conditional generative modeling all you need for decision-making?" in *International Conference on Learning Representations (ICLR)*, 2023.

[14] C. M. Jiang, A. Cornman, C. Park, B. Sapp, Y. Zhou, and D. Anguelov, "Motiondiffuser: Controllable multi-agent motion prediction using diffusion," in *IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, 2023.

[15] J. Song, C. Meng, and S. Ermon, "Denoising diffusion implicit models," in *International Conference on Learning Representations (ICLR)*, 2021.

[16] F. A. Oliehoek, S. Witwicki, and L. P. Kaelbling, "Sufficient plan-time statistics for decentralized pomdps and influence-based abstraction," *Journal of Artificial Intelligence Research*, 2021.

[17] N. Jaques, A. Lazaridou, E. Hughes, C. Gulcehre, P. Ortega, D. Strouse, J. Z. Leibo, and N. De Freitas, "Social influence as intrinsic motivation for multi-agent deep reinforcement learning," in *International Conference on Machine Learning (ICML)*, 2019.

[18] J. N. Foerster, R. Y. Chen, M. Al-Shedivat, S. Whiteson, P. Abbeel, and I. Mordatch, "Learning with opponent-learning awareness," in *International Conference on Autonomous Agents and MultiAgent Systems (AAMAS)*, 2018.

[19] C. Lu, T. Willi, C. A. S. De Witt, and J. Foerster, "Model-free opponent shaping," in *International Conference on Machine Learning (ICML)*, 2022.

[20] N. Li, D. W. Oyler, M. Zhang, Y. Yildiz, I. Kolmanovsky, and A. R. Girard, "Game theoretic modeling of driver and vehicle interactions for verification and validation of autonomous vehicle control systems," *IEEE Transactions on Control Systems Technology*, vol. 26, no. 5, pp. 1782–1797, 2018.

[21] J. F. Fisac, E. Bronstein, E. Stefansson, D. Sadigh, S. S. Sastry, and A. D. Dragan, "Hierarchical game-theoretic planning for autonomous vehicles," in *IEEE International Conference on Robotics and Automation (ICRA)*, 2019.

[22] L. P. Kaelbling, M. L. Littman, and A. R. Cassandra, "Planning and acting in partially observable stochastic domains," *Artificial Intelligence*, vol. 101, no. 1-2, pp. 99–134, 1998.

[23] Z. N. Sunberg, C. J. Ho, and M. J. Kochenderfer, "The value of inferring the internal state of traffic participants for autonomous freeway driving," in *American Control Conference (ACC)*, 2017.

[24] M. Bouton, A. Nakhaei, K. Fujimura, and M. J. Kochenderfer, "Scalable decision making with sensor occlusions for autonomous driving," *IEEE International Conference on Robotics and Automation (ICRA)*, 2018.

APPENDIX A
RELATED WORK

a) Influencing human drivers.: The closest antecedent is [4], who model a human driver as approximately optimising a learned reward and plan ego trajectories that exploit the human's best response, and its successors on active information gathering [5]. That line establishes the thesis that autonomous motion can shape human behaviour. Our departure is representational rather than conceptual: those methods rely on a structured, typically deterministic and differentiable, human response model, whereas we learn a *distribution* over joint multi-agent responses conditioned on a candidate plan, which is what lets a planner reason about disagreement between hypotheses rather than about a single best response.

b) Interaction-aware prediction.: Multimodal trajectory forecasting [6]–[9] has converged on generative models of joint futures, and conditional variants [10], [11] condition on an ego plan much as we do. These works are evaluated as predictors. We are concerned with what the conditioning is *for*: whether the induced counterfactual is causally meaningful, and whether a closed-loop agent can extract return from it.

c) Diffusion for planning and prediction.: Diffusion models have been used as trajectory planners [12], [13] and as motion predictors [14]. We use diffusion for the world model rather than the policy, because the object we need to be multimodal is the *others'* response, and we keep sampling cheap with a strided DDIM schedule [15] so that the model can be queried inside the RL loop.

d) Influence, shaping, and games.: Influence-based abstraction [16], social influence as intrinsic motivation [17], and opponent shaping [18], [19] all study agents that account for their effect on others. Game-theoretic driving models (level-k reasoning [20], Stackelberg and iterative best response [21]) assume a specified game. We assume neither a specified game nor differentiability of the other agents, and instead learn the response distribution from interaction data.

e) Belief-space planning.: Maintaining and acting on a belief over latent driver types is classical POMDP planning [22], with QMDP- and POMCP-style approximations applied to driving [23], [24]. Our belief features are a sample-based summary of a learned generative model rather than a filter over a hand-specified latent space. The summary is not a collision margin: it includes predicted neighbour waypoints and the influence feature (7), which is zero unless the forecast of y moves with the candidate plan.

APPENDIX B
IMPLEMENTATION DETAILS

A. Environment

Three SUMO maps are used: a four-way unsignalised crossing, a highway merge, and a roundabout. On the crossing the ego is inserted on the south approach with a randomised departure position and speed and traverses north; background traffic crosses east–west with Poisson arrivals whose rate is scaled by `bg_rate_scale`. A randomised warm-up of 50–110 simulation steps runs before the ego is inserted, which shifts the phase between the ego and the oncoming stream so the conflict geometry differs across episodes.

TABLE VII
NOTATION AND THE CORRESPONDING IMPLEMENTATION.

Symbol	Meaning	Code
$x_t^i = (p_t^i, v_t^i)$	pose and velocity of vehicle i	`_vehicle_snapshot`
$a_t \in \mathcal{A}$	ego acceleration, 6 discrete levels	`EnvConfig.discrete_actions`
$\theta^i \in \Theta$	latent behavioural type of vehicle i	`SocialDriverManager.types`
h_t	scene history over $H = 5$ steps	`extract_scene`
u_t	candidate ego plan over $F = 20$ steps	`synthetic_plan`
y_t	neighbours' future over F steps	`extract_future`
o_t	ego observation (ego state + $K = 5$ neighbours)	`_build_observations`
ψ_t	belief: risk, waypoints, $\Delta(u)$, intention	`BeliefEncoder`

The simulation step is $0.2\,\mathrm{s}$ and each decision repeats an action twice, giving a decision interval of $\Delta = 0.4\,\mathrm{s}$ and a horizon of 150 decisions (60 s). The ego's SUMO speed mode is set to 0, disabling safe-speed, right of way and junction-blocking checks, so collisions are possible and right of way must be resolved by the policy. Actions are accelerations drawn from $\{-4, -2, 0, 1, 2, 3\}\,\mathrm{m/s^2}$. Background vehicles use length $5\,\mathrm{m}$ (SUMO default width $\approx 1.8\,\mathrm{m}$).

a) Why the action set is zero-mean.: The natural discretisation makes the braking levels larger in magnitude than the accelerating ones, since emergency decelerations exceed any comfortable acceleration. We initially used $\{-6, -3, -1, 0, 1.5, 3\}$, whose mean is $-0.92\,\mathrm{m/s^2}$. A high-entropy policy over that set is a random walk with negative drift: the ego decelerates to a standstill within a few seconds and, since the route is roughly $200\,\mathrm{m}$ and the horizon $60\,\mathrm{s}$, can never reach the goal. The arrival reward is therefore never observed, the policy has no gradient towards crossing, and every arm converges to standing still — at 100% timeout and 0% success, with entropy collapsing well before the first success. Re-centring the action set on zero fixes this completely: the same agent reaches about 70% success within 20 iterations. We report the failure because it is a property of the action discretisation rather than of any method, it is invisible in the reward specification, and it silently equalises every arm of an ablation at zero.

Background vehicles receive a hidden type with probabilities $(0.1, 0.1, 0.8)$ over (yielder, contester, reactive). Contesting is implemented as a vehicle-type switch that removes junction-model yielding while retaining car-following safety, so a contester can collide with the ego at the junction but will not rear-end its own leader. A reactive driver commits when it first comes within $35\,\mathrm{m}$ of the conflict point on the crossing and merge ($22\,\mathrm{m}$ on the roundabout, incoming-edge allowlist), sampling from Eq. (2) with the $0.8\,\mathrm{s}$ EMA of commanded ego acceleration plus a private stubbornness offset as the logit.

A concession issues `slowDown` over 2.5 s; a contest holds or raises speed. Committing only at $\approx$ 18 m with a type switch alone left the interventional trajectory response near zero inside F steps.

b) Brake attribution.: A hard braking event is any background vehicle with $\dot{v} < -3\,\mathrm{m/s}^2$ over one decision interval. It is attributed to the ego only when the ego is within 60 m of the conflict point and the braking vehicle does *not* have a non-ego leader within 25 m. Without the leader test the count is dominated by ordinary queueing and tracks traffic density rather than ego behaviour, which makes the social reward term and the reported metric close to meaningless.

B. Data collection

Episodes are driven by a mixture: $\approx 8\%$ constant-speed, $\approx 20\%$ constant acceleration sampled from $\mathcal{A} \cup \{-4, 0, +3\} = \{-4, -2, 0, 1, 2, 3\}$, $\approx 20\%$ piecewise open-loop F-step plans (2–4 segments of length F=20), and $\approx 52\%$ gap-acceptance with per-episode aggressiveness $\mathcal{U}(-0.4, 1.0)$, accepted gap $\mathcal{U}(1.0, 2.5)$ s, and per-action noise $\epsilon \sim \mathcal{U}(0.15, 0.45)$. Background rate scale is drawn from $\mathcal{U}(0.7, 2.5)$ per episode. Only the gap arm applies ϵ-randomisation; constant and open-loop arms are already exogenous. A separate `--paired` collector forks matched $\mathrm{do}(u)$ branches from identical pre-decision histories.

Randomisation licenses $p(y \mid h, \mathrm{do}(u))$ on that mixture's support. It does not make off-support probes interventional. Default probes $\{-4, 0, +3\}$ lie on the logged support.

C. World model

The encoder maps each agent's $H = 5$ step history (5 features per step) through a two-layer MLP to a 128-d token; neighbour tokens are max-pooled for permutation invariance, the candidate plan ($F = 20$ steps, 3 features) is encoded separately, and the three are concatenated and fused to a 128-d context. The default denoiser is token-conditioned: each neighbour residual is denoised with masked joint attention rather than a flat $F \times K \times 2$ MLP. The intention head takes the context concatenated with each neighbour token.

Training uses AdamW at 3×10^{-4} with weight decay 10^{-4}, cosine annealing, batch size 256, gradient clipping at 1.0, 120 epochs, and $K = 50$ diffusion steps with a cosine $\bar{\alpha}$ schedule. The default intention-head weight is $\eta = 1.0$ (merge). On the crossing and roundabout we raise it to 4, with paired-delta weight 1.5 and a 5$\times$ weight on labelled-neighbour residual slots, so the intention loss is not drowned by curved-trajectory FDE. Supervision is masked to open-loop episodes (`interventional_intent`). Optional counterfactual Δ-loss on paired branches and interventional trajectory masking are enabled in the pipeline. The plan input is dropped with probability 0.1 so the encoder also learns a history-only branch, which guidance extrapolates from at query time. After training, a scalar guidance weight is chosen by held-out open-loop cross-entropy and written into the checkpoint. Sampling uses strided DDIM [15] with x_0 clipped to ± 4 in normalised units. When the batch enumerates counterfactual probes for one scene, the sampler reuses a single noise draw (`common_noise`) so probe differences isolate u. Normalisation constants are 30 m for history positions, 15 m/s for velocities, and $\lambda = 8$ m for the residual target.

D. Policy

PPO [3] with a two-layer 256-unit tanh actor–critic, learning rate 3×10^{-4}, $\gamma = 0.99$, GAE $\lambda = 0.95$, clip 0.2, 10 epochs per update, minibatch 256, value coefficient 0.5, gradient clipping 0.5, and 2048 environment steps per iteration for 60 iterations. The entropy coefficient is annealed linearly from 0.03 to 0.005. Observations are running-normalised per feature (Welford, frozen within an iteration); constant belief slots keep variance zero and stay at zero rather than being amplified.

The annealing schedule is not cosmetic. With a fixed coefficient of 0.01 the policy entropy collapsed from 1.78 to below 0.45 within 20 iterations, before the agent had ever completed a crossing, and every arm converged to the creeping local optimum described in Section III. The combination of the timeout penalty and sustained exploration is what makes the comparison between arms meaningful at all.

E. Belief encoder

At each step the encoder runs one batched query of constant-acceleration plans (default $\{-4, 0, +3\}$; optionally the full action set $\{-4, -2, 0, 1, 2, 3\}$), drawing S futures each with a short DDIM schedule and deterministic sampling for PPO observations. Neighbours are the K=5 nearest within 80 m. Per-probe statistics are conflict rate at $\delta = 10$ m, mean / min / spread of closest approach (scaled by 50 m), relevance-weighted intention, the predicted closest-neighbour (x, y) at mid- and end-horizon, and the mean neighbour displacement versus the hold probe ($\Delta(u)$; identically zero for `geometry` and `history`), plus five last-minus-first contrasts (38 scalars for three probes). The `geometry` ablation replaces the world model by constant-velocity neighbour roll-outs; intention and Δ are zero and risk uses the same CV core. The `kernel` arm uses that geometry and fills $P(\textsc{yield})$ / $P(\textsc{contest})$ from the analytic channel approximation.

Closed-loop runs use `plan_action` with `commit_steps` = 10 and the five arms `none` / `geometry` / `kernel` / `history` / `diffusion`.

Online, assert-minus-yield *risk* contrast is comparable for diffusion and history (ego geometry). Intention contrast is larger for diffusion but small relative to its variance. That is why a history-only or geometry policy can still be a competent risk-aware planner on $g_s(u)$ alone. The influence feature $\Delta(u)$ is the quantity those arms cannot copy; it is identically zero for `geometry` and `history` and nonzero only if $p_\phi(y \mid h, u)$ moves neighbour forecasts with the probe.

F. Qualitative setup figures

Fig. 6 shows one crossing episode as six snapshots from t_0 to t_{final}; Fig. 7a–7b show the merge and roundabout. Fig. 8 shows signed distance to the conflict on six successful episodes per map.

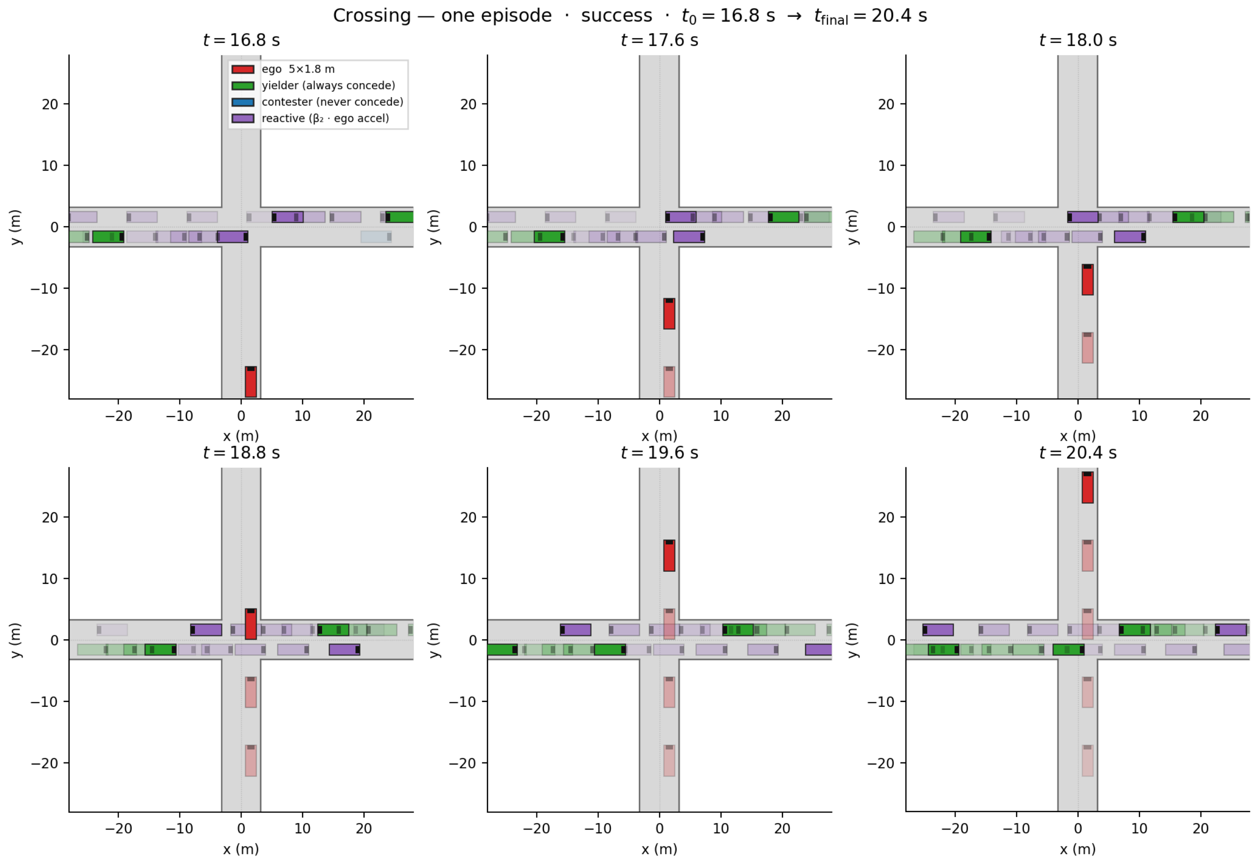


Fig. 6. Crossing running example: one successful episode, six frames from t_0 to t_{final}. Vehicles are $5{\times}1.8$ m. Red: ego; green/blue/purple: yielder / contester / reactive. Ghosted rectangles are recent history. Merge and roundabout sequences follow (Fig. 7).

G. Deciding-neighbour evaluation

The population matters, and getting it wrong hides the channel. A neighbour's intention responds to the ego's plan only if that neighbour is about to commit. Most vehicles in a scene are not: they are far from the conflict point, or have already passed it, or already decided. Their behaviour genuinely does not depend on what the ego does next, and averaging over them is averaging mostly zeros. Measured over all present neighbours, our model's probe response spans a total variation of only 0.460 (still monotone, but flattened by the zeros). Measured over the neighbours that actually commit within the plan horizon — the population the counterfactual query is about — the same model spans 0.675 against a ground-truth 0.938, and is monotone in the correct direction. We report both, because the first number is the one an incautious evaluation would produce and it is misleading. Even on the deciding population the recovered TV (0.675) is weaker than the simulator (0.938), and trajectory shifts sit near the sample spread (4.72 m): yield vs hold moves forecasts by 1.38 m, assert vs hold by 2.49 m. Directional recovery of q_ϕ is therefore not evidence that the planner receives a usable counterfactual trajectory signal.

The same consideration applies to the belief features and is not merely diagnostic. Pooling per-neighbour intentions by a plain average makes that component of ψ_t nearly constant across probes, so the policy receives no usable intention signal; we therefore pool the most dangerous partner (minimum $\Pr(\textsc{yield})$, maximum $\Pr(\textsc{contest})$ among neighbours that can reach the conflict in the query horizon). This has a corresponding cost in the supervision: the intention head is trained only on neighbours whose decision falls inside the prediction window, which is 24% of neighbour slots. Labelling every slot with the episode's eventual outcome, as would be natural, supervises the head mostly with decisions that were taken before t or long after $t+F$, and drives it to the marginal.

H. Additional tables

Table VIII and Fig. 11 vary S. $S{=}8$ is the operating point of Table III. Table X is the matched $\beta_2{=}0$ versus $\beta_2{=}2.5$ control (the $\beta_2{=}0$ rows repeat Table VI). Tables XI and XII are zero-shot evaluations with no retraining. Table XIII reports rule baselines on the crossing. Table XIV compares a joint residual to an independent per-neighbour world model; the two arms are tied on return. Table IX and Fig. 12 score interventional neighbour shift against $\mathrm{do}(u)$ replays. Main closed-loop tables for crossing, merge, and roundabout appear in Section VII-C.

TABLE VIII

SAMPLE BUDGET S FOR THE BELIEF FEATURES. REMARK 2 BOUNDS THE 95% RESOLUTION OF THE RISK ESTIMATE BY $\epsilon \approx \sqrt{\ln(40)/(2S)}$, SHOWN FOR REFERENCE. MEAN ± S.D. OVER SEEDS.

S	ϵ_{95}	Success (%)	Collision (%)	Timeout (%)	Return
1	1.37	67.60±2.80	32.20±2.75	0.20±0.60	11.04±1.02
4	0.69	67.40±2.84	32.40±3.07	0.20±0.60	10.95±1.10
8	0.48	70.00±4.00	30.00±4.00	0.00±0.00	11.97±1.49
16	0.34	68.80±3.12	31.00±3.00	0.20±0.60	11.48±1.28

TABLE IX

INTERVENTIONAL COUNTERFACTUALS ON THE UNSIGNALISED CROSSING (n=22 MATCHED SCENES). THE SAME SCENE h IS REPLAYED UNDER SEVERAL EGO PLANS $\mathrm{do}(u)$ IN THE SIMULATOR. SHIFT IS MEAN NEIGHBOUR DISPLACEMENT VS THE HOLD PLAN. A HISTORY-ONLY MODEL CANNOT MOVE ITS FORECAST WITH u.

Model	minFDE	meanFDE	Pred. shift (m)	GT shift (m)
Diffusion $p(y \mid h, u)$	2.18	32.84	1.97	1.47
History $p(y \mid h)$	1.10	20.35	0.00	1.47

TABLE X

CHANNEL-STRENGTH SWEEP. SAME WORLD MODELS (TRAINED WITH $\beta_2 > 0$); ONLY THE ENVIRONMENT KERNEL CHANGES. `HISTORY` VS `DIFFUSION` AT $\beta_2 = 0$ IS THE PLAN-CONDITIONING ABLATION WITH THE TYPE-CHANNEL REMOVED. MEAN ± S.D. OVER SEEDS.

β_2	Belief	Success (%)	Collision (%)	Timeout (%)	Return
0.00	`none`	61.80±1.40	38.20±1.40	0.00±0.00	9.10±0.54
0.00	`history`	61.80±2.09	38.00±2.37	0.20±0.60	9.00±0.83
0.00	`diffusion`	62.60±1.80	37.00±1.61	0.40±0.80	9.32±0.71
2.50	`none`	65.00±4.49	35.00±4.49	0.00±0.00	10.17±1.83
2.50	`history`	69.20±3.71	30.80±3.71	0.00±0.00	11.51±1.40
2.50	`diffusion`	70.00±4.00	30.00±4.00	0.00±0.00	11.97±1.49

TABLE XI

ZERO-SHOT SHIFT EVALUATION (HELD-OUT TYPE MIX / CHANNEL STRENGTH), MEAN ± S.D. OVER SEEDS.

Belief	Success (%)	Collision (%)	Timeout (%)	Return	Crossing steps	Induced brakes
`none` (no world model)	71.80±2.89	28.20±2.89	0.00±0.00	13.02±1.36	59.28±1.73	0.00±0.00
`history` (plan dropped)	72.20±1.89	27.60±1.96	0.20±0.60	12.92±0.81	61.85±0.78	0.00±0.00
`diffusion` (multi-hypothesis)	71.60±2.33	28.20±1.89	0.20±0.60	12.63±1.15	61.22±2.17	0.00±0.00

TABLE XII

HARDER TRAFFIC MIX (DENSER, MORE REACTIVE, STRONGER CHANNEL). NO RETRAINING. MEAN ± S.D. OVER SEEDS.

Belief	Success (%)	Collision (%)	Timeout (%)	Return	Crossing steps	Induced brakes
`none` (no world model)	62.20±3.74	37.80±3.74	0.00±0.00	8.88±1.54	58.80±1.90	0.00±0.00
`history` (plan dropped)	65.40±3.90	33.80±3.40	0.80±1.33	9.90±1.68	62.08±2.13	0.00±0.00
`diffusion` (multi-hypothesis)	64.80±5.74	35.20±5.74	0.00±0.00	9.75±2.37	61.38±2.48	0.00±0.00

TABLE XIII

RULE-BASED BASELINES ON THE UNSIGNALISED CROSSING (SAME CHANNEL AND TYPE MIX AS THE PPO ARMS; 80 EPISODES).

Policy	Success (%)	Collision (%)	Timeout (%)	Return
`gap` (time-gap acceptance)	73.75	26.25	0.00	6.02
`constant` (always-go)	70.00	30.00	0.00	13.11

TABLE XIV

INDEPENDENT PER-NEIGHBOUR DIFFUSION WORLD MODEL VERSUS THE JOINT RESIDUAL ON THE CROSSING, MEAN ± S.D. OVER n=10 SEEDS. THE TWO ARMS ARE TIED ON RETURN.

Belief	Success (%)	Collision (%)	Timeout (%)	Return	Crossing steps	Induced brakes
`diffusion` (joint)	70.00±4.00	30.00±4.00	0.00±0.00	11.97±1.49	62.13±2.97	0.00±0.00
`independent` (per neighbour)	71.60±4.72	28.40±4.72	0.00±0.00	12.44±1.76	64.12±3.30	0.00±0.00

I. Learning curves

Learning curves deferred from Section VII-C are collected here. Fig. 13 shows PPO return on the crossing under the operating point (`commit_steps`=10). Fig. 14 repeats the same curves on the merge and roundabout. On the crossing, `diffusion` tracks `geometry`; on the merge it pulls away;

on the roundabout the arms stay overlapped, matching the closed-loop tables in Section VII-C.

J. Compute

World-model training (120 epochs, batch 256) takes about 20 minutes on one NVIDIA GPU. Each PPO run is 60 iterations $\times$ 2048 environment steps; wall-clock is dominated by SUMO and is roughly 6–8 hours per seed depending on the belief encoder. The default suite targets 10 seeds $\times$ 5 arms $\times$ 3 scenes; merge seed counts are incomplete, as noted in Table IV. Optional ablations (matched β_2=0, S-sweep, independent WM, harder mix / shift, rule baselines) use the same crossing operating point. All experiments were run on a local 4-GPU workstation. Preliminary failed runs (skewed action set, missing timeout penalty) are not included in the reported seed counts.

APPENDIX C
REWARD DESIGN AND FAILURE MODES

Two terms deserve comment. The social term is what separates negotiating from bullying: forcing right of way is otherwise free, and an unpenalised agent converges to it. Crucially, a braking event is attributed to the ego only when the ego is near the conflict point *and* the braking vehicle is not simply following a close leader; without that test the statistic is dominated by ordinary queueing and measures traffic density rather than ego behaviour. The liveness term is equally load-bearing for a less interesting reason. Under Eq. (3) without w_{to}, creeping at $v^\star = w_\tau v_{\max}/w_p \approx 4.2\,\text{m/s}$ exactly cancels the time penalty, making "never cross" a stable local optimum that dominates any exploratory attempt to cross; in preliminary runs every method collapsed onto it. We report this because it is a property of the reward, not of any method, and it would silently invalidate a comparison between methods.

APPENDIX D
IDENTIFICATION DISCUSSION

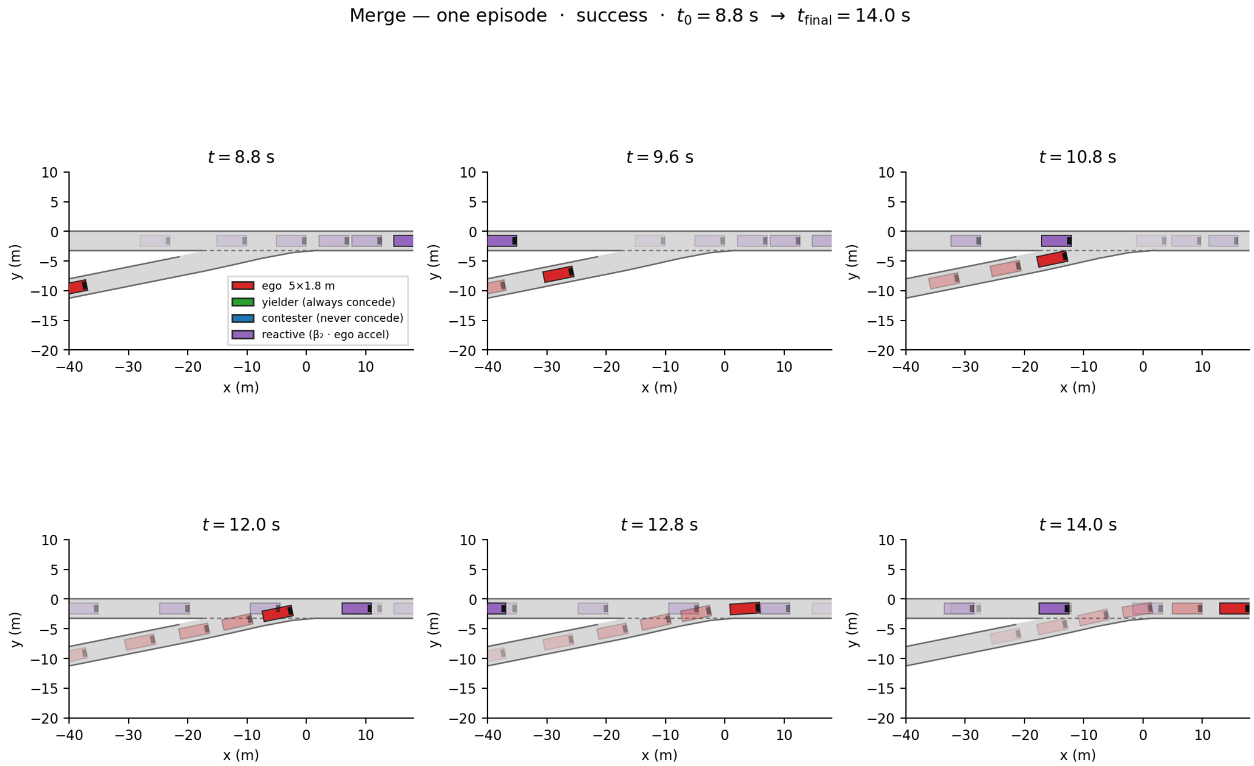
This is the step at which conditional-prediction-as-counterfactual claims usually fail, so we state it explicitly. Suppose the data are generated by a reactive controller $u = \pi_b(h)$. Then u carries no variation given h, $p(y \mid h, u) = p(y \mid h)$, and evaluating the model at a plan the controller would not have chosen is extrapolation, not intervention. Worse, if the controller has episode-level parameters (aggressiveness, accepted gap) that also influence y, those parameters are unobserved confounders and the learned conditional is biased for the interventional quantity $p(y \mid h, \mathrm{do}(u))$.

We break both problems at the source, in the behaviour policy rather than in the estimator. Episodes are a mixture: $\approx$ 8% constant-speed, $\approx$ 20% constant acceleration drawn from the discrete action set $\mathcal{A} = \{-4, -2, 0, 1, 2, 3\}$, $\approx 20\%$ piecewise open-loop F-step plans (identifying $\mathrm{do}(u_{1:F})$, not only one-step u_t), and $\approx$ 52% gap-acceptance with per-action $\epsilon \sim \mathcal{U}(0.15, 0.45)$ drawn per episode. The random and open-loop components are exogenous by construction, so conditioning on (h, u) identifies $p(y \mid h, \mathrm{do}(u))$ on the support of that mixture — not off it. Matched-branch (`--paired`) collection additionally forks the same history under several constant-acceleration probes so counterfactual Δ-losses can supervise response differences. The planner's default probes $\mathcal{U} = \{-4, 0, +3\}$ are constant-acceleration roll-outs of on-support actions (yield / hold / assert). Section VII-B checks that the probe response has the kernel's *sign*; that is directional recovery, not identification of ρ.

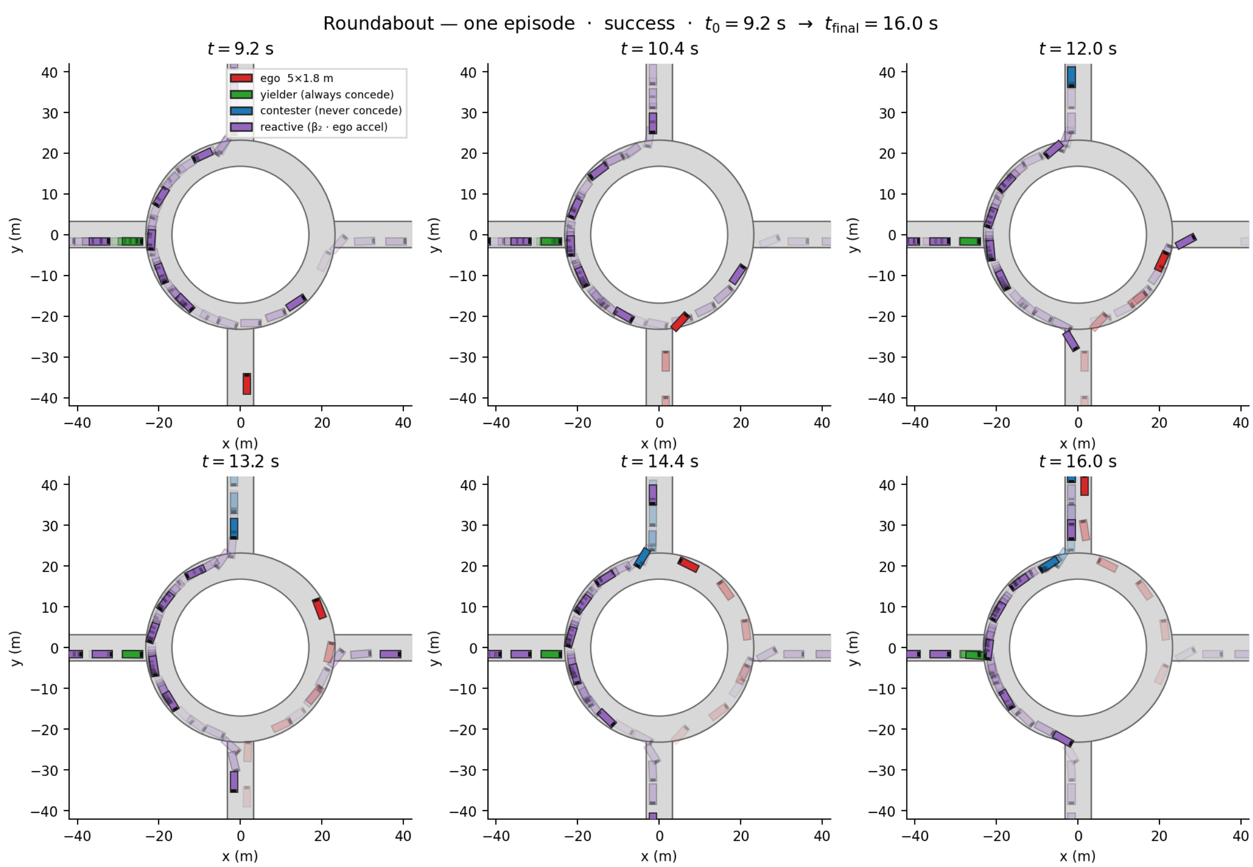
APPENDIX E
BELIEF-FEATURE CONSTRUCTION DETAILS

Assert-minus-yield contrasts on risk, margin, spread, intention and Δ bring the vector to 38 scalars. Relevance pooling is a softmax over predicted closest approach, not identification of "deciding" vehicles and that label is evaluation-only.

A `geometry` ablation replaces the learned world model by constant-velocity neighbour roll-outs against the same probe ego plans, so conflict risk is pure engineered geometry and $\Delta \equiv 0$. A `kernel` arm uses that same geometry and fills the intention slots from an analytic approximation to the simulator channel (marginalising stubbornness), rolled forward under the probe's EMA. It is a strong engineered baseline for what an accurate channel feature adds over `geometry`; it is *not* an exact matched-branch oracle. Exact branch evaluation is reported separately when available.



(a) Merge.



(b) Roundabout.

Fig. 7. One successful episode per map, six frames from t_0 to t_{final}, matching Fig. 6. Vehicles are 5×1.8 m. Red: ego; green/blue/purple: yielder / contester / reactive.

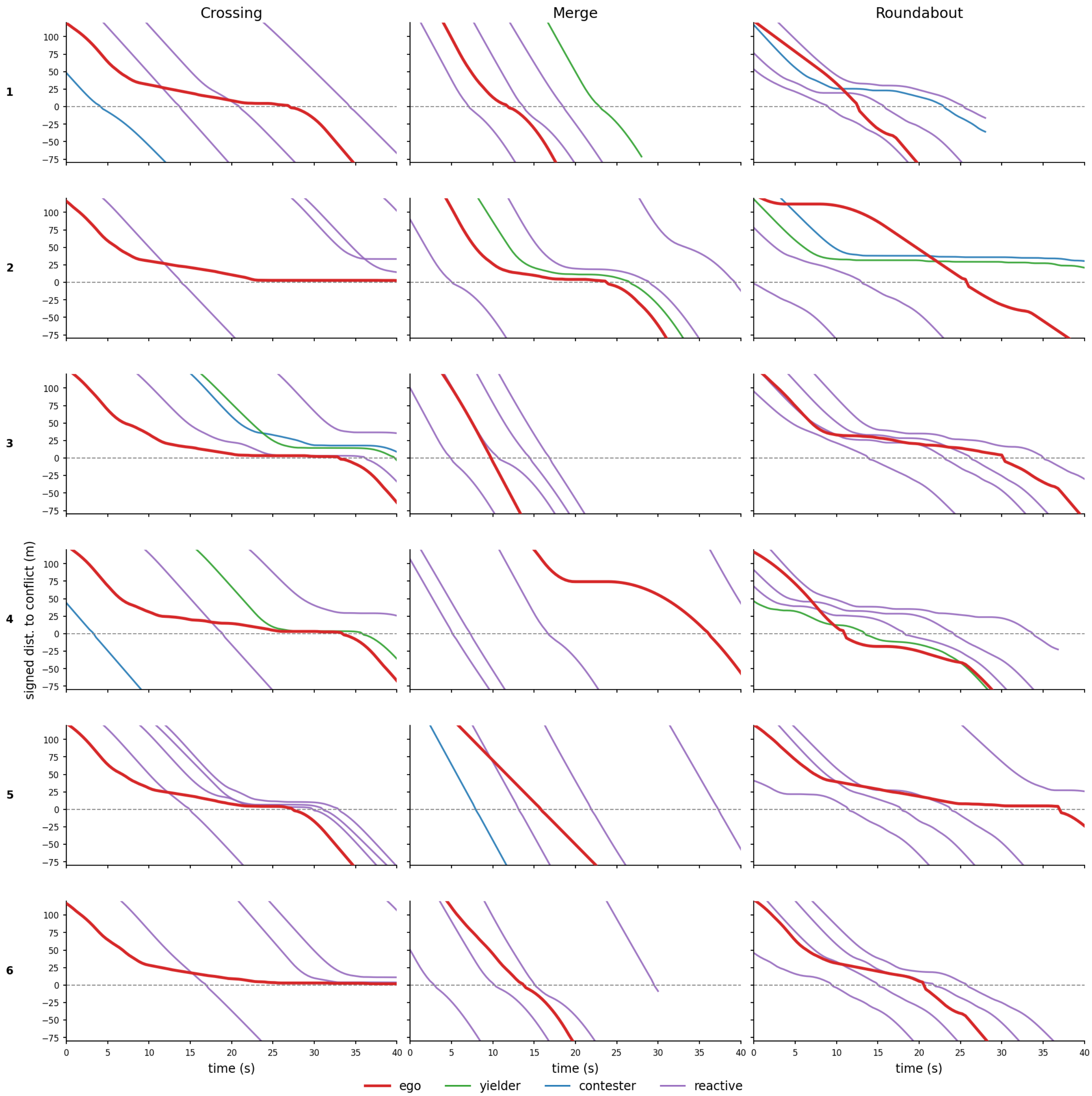


Fig. 8. Signed distance to the conflict for six successful episodes per map (rows). Red: ego; green/blue/purple: yielder / contester / reactive. $d=0$ is the conflict. A plateau just above zero is waiting; a crossing of the dashed line is clearance.

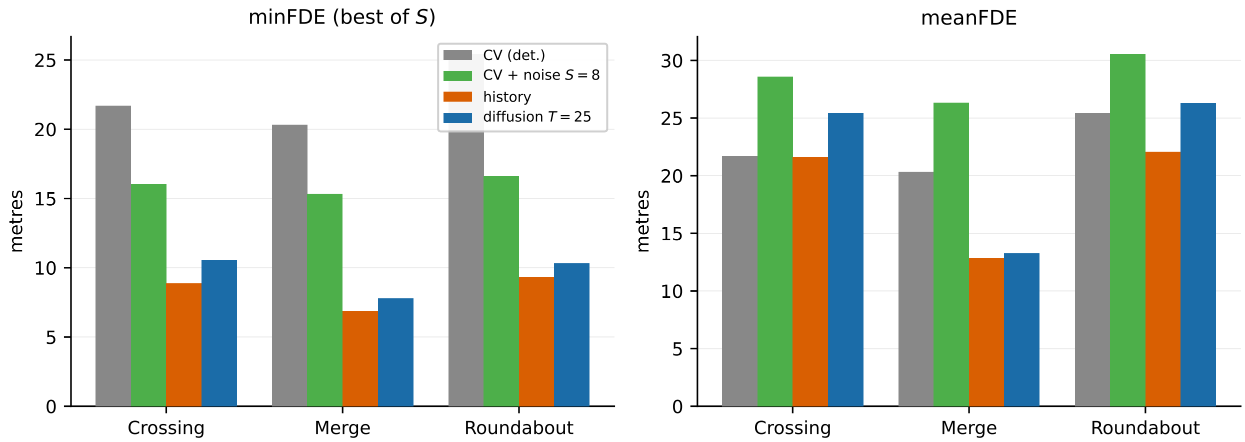


Fig. 9. Held-out displacement error on all three maps. minFDE is a best-of-S statistic; meanFDE is the proper comparison to deterministic constant velocity. History is competitive on minFDE; diffusion meanFDE does not dominate CV on the crossing.

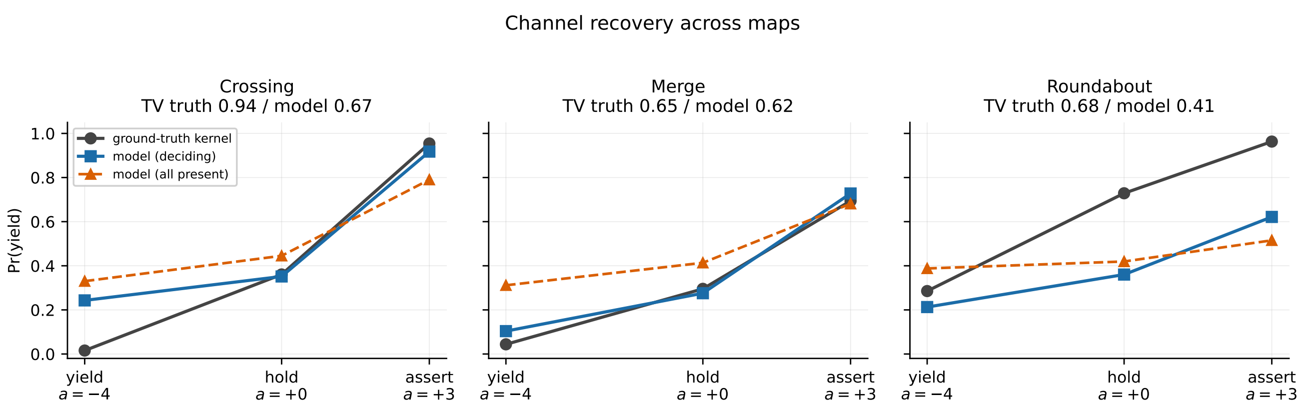


Fig. 10. Same intention-head probe as Fig. 2, on all three maps. The deciding-neighbour population is the relevant one. Merge recovers almost all of the kernel TV; the roundabout recovers the direction but not the magnitude.

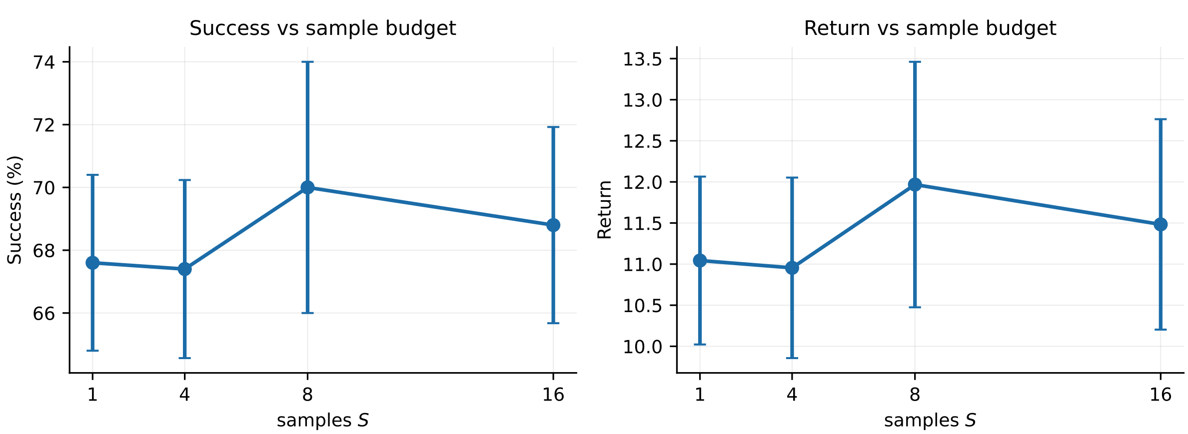


Fig. 11. Closed-loop success and return versus belief sample budget S on the crossing. S=8 is the operating point; S=1 is already close.

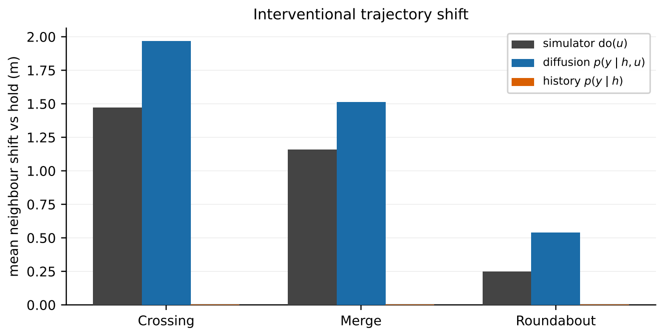


Fig. 12. Interventional neighbour shift versus the hold plan. History cannot move with u (shift identically zero). Diffusion overshoots the simulator slightly but has the correct sign on every map.

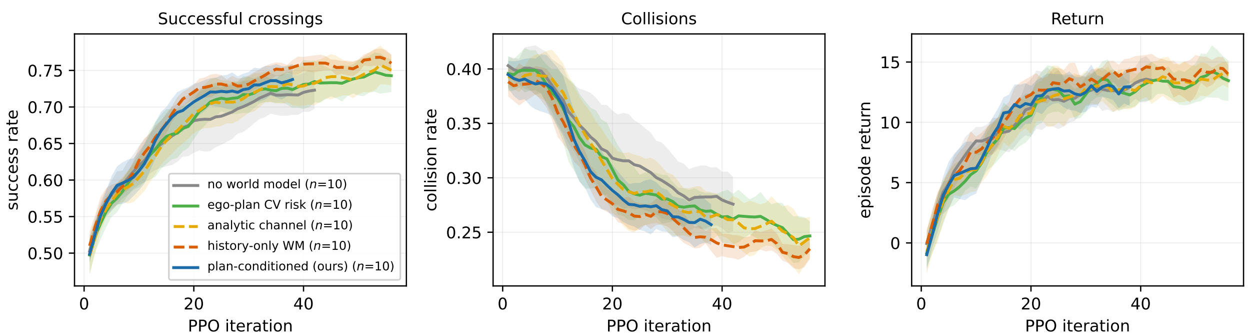


Fig. 13. Learning curves on the crossing (mean ± s.d. over seeds). Under `commit_steps`=10, `diffusion` tracks `geometry`; `history` and `kernel` are statistically tied with geometry.

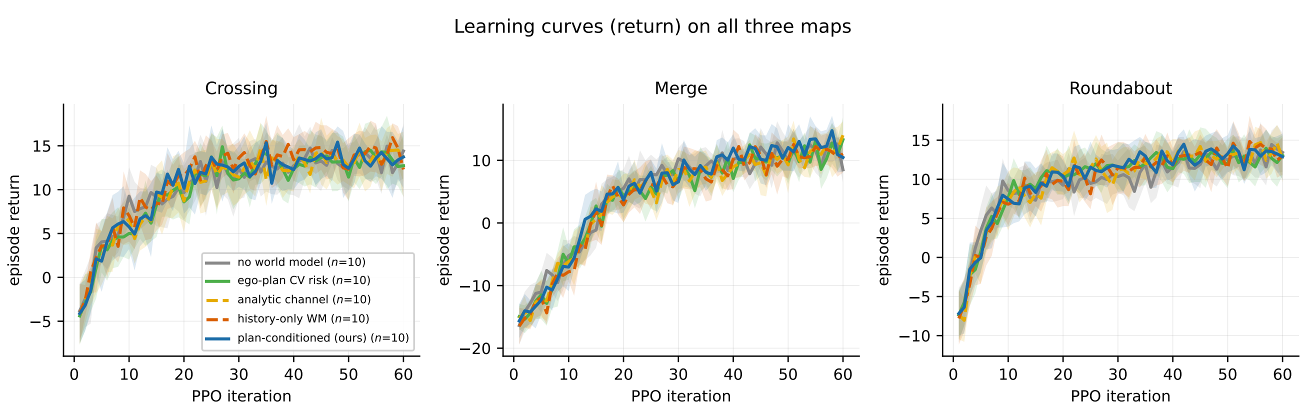


Fig. 14. Return learning curves on all three maps (mean ± s.d. over seeds). Diffusion pulls away only on the merge; on the crossing and roundabout the arms stay overlapped.